A QUASI-COMPACT QUASI-SEPARATED MORPHISM OF ALGEBRAIC STACKS THAT DOES NOT PRESERVE D_QCOH UNDER DERIVED PUSHFORWARD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The middle context span is the statement environment of lemma-is-limit only; its proof is not part of the delivery. The two 'insert future reference here' placeholders flag standard group-cohomology references and sit in the construction layer.

Editorial note. The label lemma-push-not-OK names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 4920-4986: k of characteristic p > 0, S = Spec(k[x]), X = [S/G] with G = Z/pZ acting trivially (Examples of Stacks, X-mod-G-group; Criteria for Representability, BG-algebraic); f: X -> S is quasi-compact and quasi-separated so Rf_{QCoh,*}: D^+_QCoh(X) -> D^+_QCoh(S) exists (Derived Categories of Stacks, derived-direct-image-quasi-coherent); RGamma(X, O_X) is computed by the Cech complex k[x] -> Map(G, k[x]) -> Map(G^2, k[x]) -> ... (Cohomology of Stacks, smooth-covering-compute-cohomology; S x_X ... x_X S = S x G^p affine); this is the group cohomology of G acting trivially on k[x] (the author's two 'insert future reference here' placeholders flag the standard references, kept verbatim per the L48293 precedent); the cohomology of the cyclic group G on k[x] is one copy of k[x] in each degree >= 0, giving Rf_*O_X = direct sum over n >= 0 of O_S[-n]; E = direct sum over m >= 0 of O_X[m] is an object of D_QCoh(O_X)
- Context span 4989-5005 (lemma-is-limit statement): for K with locally quasi-coherent cohomology sheaves satisfying the flat base change property there is a distinguished triangle K -> prod tau_{>= -n} K -> prod tau_{>= -n} K -> K[1], i.e., K is the derived limit of its canonical truncations; the delivered card L48295 (sum-is-product) gives direct sum = product in D_QCoh for such K
- Context span 5041-5057: since a quasi-coherent module is locally quasi-coherent with the flat base change property (Sheaves on Stacks, quasi-coherent) we get E = prod over m >= 0 of O_X[m]; since cohomology commutes with limits, Rf_*E = prod over m >= 0 of (direct sum over n >= 0 of O_S[m-n]); the degree-0 cohomology sheaf is an infinite product

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of copies of O_S, which is not even locally quasi-coherent, so this complex is not an object of D_QCoh(O_S)

Source reference. Original lemma 06PL: sources/stacks-chow/tags.txt, source lines 8575–8575. Prerequisite pointer invoked at 5077; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0371: sources/stacks-chow/tags.txt, source lines 4184–4184. Prerequisite pointer invoked at 5075; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 06Z0: sources/stacks-chow/tags.txt, source lines 8904–8904. Prerequisite pointer invoked at 5096; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 0762: sources/stacks-chow/tags.txt, source lines 9151–9151. Prerequisite pointer invoked at 5145; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 0784: sources/stacks-chow/tags.txt, source lines 9224–9224. Prerequisite pointer invoked at 5107; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 07BC: sources/stacks-chow/tags.txt, source lines 9337–9337. Prerequisite pointer invoked at 5091; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 06WJ: sources/stacks-chow/tags.txt, source lines 8818–8818. Prerequisite pointer invoked at 5143; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 06WK: sources/stacks-chow/tags.txt, source lines 8819–8819. Prerequisite pointer invoked at 5158; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Let k be a field of characteristic $p > 0$. Let $S = \mathrm{Spec}(k[x])$. Let $G = \mathbf{Z}/p\mathbf{Z}$ viewed either as an abstract group or as a constant group scheme over S. Consider the algebraic stack $\mathcal{X} = [S/G]$ where G acts trivially on S, see Examples of Stacks, Remark 0371 and Criteria for Representability, Lemma 06PL. Consider the structure morphism

$$f : \mathcal{X} \longrightarrow S$$

This morphism is quasi-compact and quasi-separated. Hence we get a functor

$$Rf_{QCoh,*} : D^+_{QCoh}(\mathcal{O}_\mathcal{X}) \longrightarrow D^+_{QCoh}(\mathcal{O}_S),$$

see Derived Categories of Stacks, Proposition 07BC. Let's compute $Rf_{QCoh,*}\mathcal{O}_\mathcal{X}$. Since $D_{QCoh}(\mathcal{O}_S)$ is equivalent to the derived category of $k[x]$-modules (see Derived Categories of Schemes, Lemma 06Z0) this is equivalent to computing $R\Gamma(\mathcal{X}, \mathcal{O}_\mathcal{X})$. For this we can use the covering $S \to \mathcal{X}$ and the spectral sequence

$$H^q(S \times_\mathcal{X} \ldots \times_\mathcal{X} S, O) \Rightarrow H^{p+q}(\mathcal{X}, \mathcal{O}_\mathcal{X})$$

see Cohomology of Stacks, Proposition 0784. Note that

$$S \times_{\mathcal{X}} \ldots \times_{\mathcal{X}} S = S \times G^p$$

which is affine. Thus the complex

$$k[x] \to \mathrm{Map}(G, k[x]) \to \mathrm{Map}(G^2, k[x]) \to \ldots$$

computes $R\Gamma(\mathcal{X}, \mathcal{O}_{\mathcal{X}})$. Here for $\varphi \in \mathrm{Map}(G^{p-1}, k[x])$ its differential is the map which sends $(g_1, \ldots, g_p)$ to

$$\varphi(g_2, \ldots, g_p) + \sum_{i=1}^{p-1} (-1)^i \varphi(g_1, \ldots, g_i + g_{i+1}, \ldots, g_p) + (-1)^p \varphi(g_1, \ldots, g_{p-1}).$$

This is just the complex computing the group cohomology of G acting trivially on $k[x]$ (insert future reference here). The cohomology of the cyclic group G on $k[x]$ is exactly one copy of $k[x]$ in each cohomological degree ≥ 0 (insert future reference here). We conclude that

$$Rf_*\mathcal{O}_{\mathcal{X}} = \bigoplus\nolimits_{n \geq 0} \mathcal{O}_S[-n]$$

Now, consider the complex

$$E = \bigoplus\nolimits_{m \geq 0} \mathcal{O}_{\mathcal{X}}[m]$$

This is an object of $D_{QCoh}(\mathcal{O}_{\mathcal{X}})$. We interrupt the

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Let K be an object of $D(\mathcal{O}_{\mathcal{X}})$ whose cohomology sheaves are locally quasi-coherent (Sheaves on Stacks, Definition 06WJ) and satisfy the flat base change property (Cohomology of Stacks, Definition 0762). Then there exists a distinguished triangle*

$$K \to \prod\nolimits_{n \geq 0} \tau_{\geq -n} K \to \prod\nolimits_{n \geq 0} \tau_{\geq -n} K \to K[1]$$

in $D(\mathcal{O}_{\mathcal{X}})$. In other words, K is the derived limit of its canonical truncations.

We continue our discussion. Since a quasi-coherent module is locally quasi-coherent and satisfies the flat base change property (Sheaves on Stacks, Lemma 06WK) we get

$$E = \prod\nolimits_{m \geq 0} \mathcal{O}_{\mathcal{X}}[m]$$

Since cohomology commutes with limits we see that

$$Rf_*E = \prod\nolimits_{m \geq 0} \left(\bigoplus\nolimits_{n \geq 0} \mathcal{O}_S[m - n] \right)$$

Note that this complex is not an object of $D_{QCoh}(\mathcal{O}_S)$ because the cohomology sheaf in degree 0 is an infinite product of copies of $\mathcal{O}_S$ which is not even a locally quasi-coherent $\mathcal{O}_S$-module.

Lemma 0.2. *A quasi-compact and quasi-separated morphism $f : \mathcal{X} \to \mathcal{Y}$ of algebraic stacks need not induce a functor $Rf_* : D_{QCoh}(\mathcal{O}_{\mathcal{X}}) \to D_{QCoh}(\mathcal{O}_{\mathcal{Y}})$.*

Proof. See discussion above. □